

RDC / FRDC Confidentiality Handbook for Economic Microdata

Data Access Division

This document is non-confidential and can be shared with data users outside of the RDC / FRDC. Updates to this document are made regularly by the DAD Vetting Committee.

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Summary Checklist for Vetting Considerations for Economic Data

	Vetting Considerations for Economic Data Requests	Section(s) to consult
✓	Output is consistent with the approved proposal and evaluation	• Analytic Outputs Must Be Consistent with the Approved Proposal and Evaluation
✓	Descriptives are limited and support analytical output	• What Is “Analytical Output”? • Intermediate versus Final Outputs • Summary Statistics • Residual Disclosure • Vetting Considerations for Counts and Summary Statistics
✓	All rounding and minimum count guidelines have been applied	• Rounding • Minimum counts / Sample Sizes • Derived Statistics • Quantiles • Vetting Considerations for Counts and Summary Statistics
✓	Analysis-specific rules have been applied	• Longitudinal Analysis • Custom Aggregations of Variables • Graphs, Figures and Maps • Linear combinations of variables
✓	The unique unit of analysis is correctly identified	• Minimum Counts / Sample Sizes • Vetting Considerations for Counts and Summary Statistics
✓	Sensitivity Tests have all passed	• Precision Measures • Sensitivity Testing • Sensitivity Marco for Economic Data • Appendix A – General Description of the Sensitivity Macro
✓	Outputs that did not pass were redesigned. Suppression is not permitted	• Suppression
✓	Guidelines for model outputs have been followed	• Vetting Considerations for Model-Based Outputs • Checklist for Vetting Models
✓	The vetting request form has been filled out correctly	• Vetting Requests • Appendix B: DAD Confidentiality Vetting Request Form for Business Data
✓	There is no residual disclosure within the request or between other vetting requests	• Residual Disclosure • Vetting Output from Extensions / Revisions of an Ongoing Contract • Disclosure by repeated differencing / Nested Models

Preface

The process of creating analytical outputs from the economic microdata collected by Statistics Canada is authorized under the Statistics Act (R.S.C., 1985, c. S-19). The Statistics Act mandates Statistics Canada to collect, analyze and publish information on and about Canadians while simultaneously protecting the confidentiality of respondents. This document deals with the latter responsibility.

For the purposes of this document, all **data users, or researchers are “deemed employees” of Statistics Canada**. This means that all data users will, with respect to data confidentiality, (1) sign an Oath, (2) comply with Statistics Canada policies and protocols, and (3) access and use data as strictly outlined in their Data Access Agreement as explained in the Researcher Guide - Data Access Division document. All data users are also responsible for creating and submitting what are considered “Safe Outputs” for confidentiality vetting and release from **Statistics Canada Secure Access Points¹ (SAPs)**.

The primary goal of the vetting process (hereafter known as “vetting”) is to protect the confidentiality of the data and of the information provided by the respondents. As part of this, all data users will (1) complete vetting training and will have access to vetting training materials, and (2) follow the vetting guidelines to create safe outputs. Statistics Canada Analysts will review (vet) all output before it is released from a secure environment, to ensure the ongoing protection of the data and to maintain the trust of Canadians.

All of us play a role in maintaining the confidentiality of the data.

This document is not meant to be a guide on statistical procedures for analyzing data. It focuses on the considerations for vetting output for compliance with confidentiality guidelines and on what documentation is required for output to be properly vetted to maintain confidentiality.

This document is updated regularly by the DAD Vetting Committee.

¹ **Secure Access Point (SAP)**: a location specified in the Data Access Agreement in which a Deemed Employee can use Protected Information and that meets Statistics Canada’s departmental security standards for data access based on the identified level of risk.

The Vetting Approach for Economic Data

The vetting approach always involves risk mitigation rather than absolute avoidance. It seeks to maximize the usefulness of Statistics Canada's economic microdata for research purposes while simultaneously minimizing the risk of disclosure of confidential information. If these goals conflict, and where judgement must be applied, reducing the risk of disclosure takes precedence.

The vetting approach for economic data must recognize two specific risks. The first is the risk of releasing analytical output that reveals specific information about a business institution, which leads to that business institution being identified within analytical outputs. For example, aggregating information from several establishments that are part of a single enterprise yields information about that enterprise, even though there appears to be many different businesses; another example is when "residual disclosure" occurs, where multiple outputs are combined that lead to the identification of a business institution. The second risk is that contextual or behavioural information about a business institution can lead to its identity being inferred from an output (known as attribute disclosure), especially if the analytical output involves information of a sensitive nature (e.g., wages paid to employees). For example, in the case of sums for variables confined to small industries and/or geographic areas there can be a considerable amount of information in the filters applied to select the units within the analysis that aids with estimating the values of a particular business units.



Data users are responsible for learning the vetting rules and vetting process at the beginning of the project, as well as ensuring that all applicable vetting rules are applied to all output for release.

Vetting Guidelines for Economic Microdata

The vetting of economic microdata can be complex. In the following sections, general guidelines and considerations for vetting analytical output derived from economic data are presented. The vetting guidelines for the specific data being used are contained in the confidential document "DAD Vetting Guidelines for Economic Data" which can be obtained from your local Statistics Canada Analyst.

Outputs Must Be Consistent with the Approved Proposal and Evaluation

Analysts will be checking that the output being requested for vetting is consistent with what was specified in the project proposal. If the Analyst is unsure if the requested output is consistent with the proposal they may send the output to the DAD Vetting Committee for a recommendation, which could increase the time required to release the output.

Substantive deviations from the proposal should be discussed with the Analyst – changes to variable coding or types of predictor variables are generally not considered substantive. If there is a change in the outcomes being examined (e.g., changes in the industries being examined) this may require an amendment to the project proposal and data users are encouraged to discuss this with the Analyst.

Data users should review the Subject Matter evaluation of their proposal with the Analyst as these may have some important information related to vetting the output. If the evaluation was not sent to data users, they can request to review it with the Analyst.

What Is “Analytical Output”?

“Analytical Output” refers to each separate analysis (as defined by the data user) in a vetting request.

The primary type of analytical output that will be released from the RDCs / FRDC with the economic microdata will be model output. Descriptive output will be limited to those that would typically be found in a basic table in Journal articles, including overall high-level single-variable descriptions of the sample (e.g., counts of businesses in the sample) and variables to support the models such as counts, proportions, averages, and quantiles. Only summary counts / descriptives at a high level describing the sample are allowed for vetting requests; 2-way tables (e.g., means of revenue by industry code and firm size) can only be released if they are (1) specifically requested in the proposal or are within the scope of the proposal, (2) they are supporting output for a model or will support a model in the near future, and (3) they are a high aggregate level (i.e., by province, or gender, or 2-digit NAICS). Data users will be expected to use [publicly available tables](#) or [request custom tabulations](#) for more detailed descriptive statistics.

Data users of economic data should also be aware that applying successive filters to a set of model outputs to build a detailed descriptive table from model sample sizes is not allowed. For example, a researcher could stratify regression models by every combination of business size and industry type, but this will yield a crosstabulation table of counts for business size and industry type using the model sample sizes that is not releasable. However, the model coefficients may be releasable without the sample sizes if all other applicable vetting rules are met.

A model with independent variables A and B and a model with independent variables A, B, and C are considered two separate pieces of analytical output, even if they have the same dependent variable and sample size. Data users should be aware that this view of what is considered analytical output can have a significant impact on the amount of time required to process a vetting request, since every piece of analytical output must be checked individually.



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Intermediate versus Final Outputs

Intermediate outputs for projects using economic data are not permitted for release. Residual disclosure is most likely to occur as adjustments to the sample occur over the course of the project and this may prevent Statistics Canada from releasing the adjusted summary statistics in later requests. However, it is understood that data users may need to evaluate the progress of their project prior to submitting the final output for vetting. Data users should discuss with the Analyst options for releasing non-numerical versions of intermediate output – for example, a summary table indicating which coefficients in a regression model were positive / negative and significant / not significant could be released prior to the final version of the output. Rounding in isolation is not an acceptable solution for preventing residual confidentiality issues with intermediate outputs.

Vetting Outcomes

There are 4 general outcomes for a vetting request:

1. The output meets all the applicable vetting rules and has no residual issues with any output released for the project. This output is releasable.
2. The output (in whole or in part) does not meet all the applicable vetting rules but the rules can be met with some modification to the variables or analyses. This could involve needing to recode some variables or adjust analyses so that the output that is not releasable can be made releasable.
3. The output (in whole or in part) does not meet all the applicable vetting rules and there is no way to adjust the analyses to make the output releasable. This can happen – for example, a dichotomous variable with counts that do not meet the minimum threshold cannot be collapsed any further; a survival curve can require so much grouping of observations and smoothing that the resulting curve has no meaning.

Vetting Considerations for Outputs

Rounding

Rounding of counts, in addition to minimum counts, is required for economic data outputs as it gives visible indication that steps are being taken to mitigate potential identity disclosure and offers more protection against residual disclosure. Rounding is applied after all other applicable vetting rules are applied, and the appropriate rounding base can be found in the vetting rules for economic data. All counts (e.g., number of enterprises, total number of employees by industry) must be rounded according to the rounding base specified in the vetting rules for economic data. **Means for continuous variables do not require rounding, whereas totals of continuous variables must be calculated by multiplying the rounded count by the unrounded mean**

There are many rounding approaches that are available to data users (deterministic, random, additive, and controlled), and the rules for economic data will specify which rounding method is required; if no specific method is advised then the choice of rounding method is left up to the data user, with the condition that the rounding method is applied consistently throughout the project. The Analyst can provide some guidance and documentation on rounding procedures, and some of them are described in

the DAD Vetting Handbook (available from the Analyst). The use of random or additive controlled rounding for counts or descriptives is strongly recommended.

Summary Statistics

Summary and tabular output are restricted for economic data being accessed through Research Data Centres. Access to economic data is granted to facilitate analytic research projects, not for data users to create custom tabular output. Tabulations are allowed to describe the data set being used in a research paper and to motivate the analysis (see the above section on [Analytical Output](#)). These tabulations are general, high-level descriptives that support the analysis. These are not cross-tabulations that offer analytic utility. Below are some examples of descriptive tables that may be requested for vetting. Examples A and B are at a high level (using NAICS codes for an entire fiscal year for all of Canada). However, Example C may or may not be at high enough level (using private vs public vs unclassified businesses) to be released from an SAP, regardless of meeting other vetting rules, depending on the level of detail in the context of the approved proposal and corresponding model output that this table would be supporting. Example C would need to be evaluated in more detail by the Analyst.

• Example A Mean Revenue and Median Expenditure by NAICS code for Entire 2002 Fiscal Year			• Example B Count of businesses by Number of employees and NAICS code for Entire 2002			• Example C Count of businesses by Enterprise Status and Employee Count for Entire 2002 Fiscal Year		
NAICS Code	Mean Revenue	Median Expenditure	NAICS code	< 1000 staff	1000+ staff	Status	< 1000 staff	1000+ staff
20013			20013			Private		
20104			20104			Public		
22005			22005			Not Classified		

Example A and B are marked with a green checkmark, indicating they are acceptable. Example C is marked with an orange checkmark and a question mark, indicating it requires further evaluation.

Minimum Counts / Sample Sizes

For all output, there is a minimum number of unique identifiable units (M) that must be present for the output to be considered for release (this minimum is considered confidential and cannot be shared outside of the SAP). For proportions to be releasable, the numerator, denominator, and the difference between the numerator and denominator must meet the minimum count requirement to protect against residual issues from any omitted categories. For model-based output, several tabulations may be required to show that all minimum count requirements are met (see the later section on [Vetting Considerations for Model-Based Outputs](#)).

If a count does not meet the minimum threshold, data users must redefine their sample or combine / collapse variable categories within the sample so that the counts meet the minimum thresholds. Minimum and maximum values are not releasable, even if these values meet the vetting requirements for each summary statistic. The section on [residual disclosure](#) has more information on minimums for non-response.

Some datasets (e.g., CEEDD, WES) have multiple layers of units (owners, businesses, enterprises, establishments and/or employees), thus containing a **hierarchical structure** (for example, an enterprise or a business may have multiple establishments). Data users may be interested in the relation between

employers and employees, or owners and their businesses, etc. In such cases, it is important for both data users and Analysts to understand the structure of the data and how the analysis relates to that structure.

Data users are strongly encouraged to read through available documentation on the economic data being used to understand the hierarchy. **This assessment of counts at a hierarchical level will likely reduce the observed cell count to a newer (and usually lower) “actual” cell count; this new count is what will be used for assessing the minimum count rules.** The need for the application of this hierarchy with particular data sets will be explicitly mentioned in the appropriate vetting sections below. Further, given the nature of vetting economic data, meeting the minimum count threshold does not guarantee the release of the output as all output must pass the [sensitivity tests](#) described below.



Throughout this document, **the unique data unit or unit of analysis** used for confidentiality rules is generally what is **defined as a single "unit" in the universe of data points being considered -- the owner, the business (an enterprise, not an establishment), or the employee.** For example, if the universe of data points is all owners of a business (regardless of the type of business), then minimum counts will generally be applied to the number of owners, not the number of employees nor the number of businesses. Unless otherwise specified, thresholds and rounding bases apply to whichever unit of reference is required (i.e., owner, business, or employee).

Linear Combinations of Variables

Linear combinations are assessed on the aggregated variable (i.e., meeting the minimum thresholds and passes all applicable sensitivity tests) and not the individual components. An example of a linear combination is:

$$\text{Total Income} = \text{Social Assistance} + \text{Provincial Transfer Payments} + \text{T4 Income}$$

In this example, only the variable Total Income is assessed. If in the future the researcher wishes to release information on the individual components, they must be assessed, and caution must be taken to ensure any excluded components are not revealed (and if so that they are assessed and pass all applicable sensitivity tests).

Derived Statistics

Derived statistics such as proportions, level and percent changes, and ratios, are releasable if each component is releasable. Some weighted averages may be the ratio of two totals – in this situation the weighted average can be released if both the numerator and denominator are releasable. Moving averages and other longitudinal statistics calculated over multiple time periods (e.g., years, months) must pass the vetting guidelines for each time period included in the vetting request. For averages that are the ratio of two totals (e.g., the average labour productivity defined as labour output over employment) or a ratio of two totals multiplied by a value, that average can be released if both the numerator and denominator pass the vetting guidelines.

For proportions, the following three-tiered approach can be used to determine the appropriate rounding strategy, depending on the number of units in the denominator:

Denominator	Calculate the proportion based on	Decimal places
> 5,000	Unrounded numerator and denominator	3 (or 0.1% for percentages)
500 to 4,999	Unrounded numerator and denominator	2 (or 1% for percentages)
< 500	Rounded numerator and denominator	As many as desired

Quantiles

Quantiles are exact data points and must have the minimum number of unique units (M) below or at the quantile of interest and the minimum number of unique units above or at the quantile of interest. For example, for the median income to be released, there should be at least M units with an Income below or equal to the median income (50% point) and at least M units with an Income above or equal to the median income (50% point). If more than one quantile is produced, then there must also be at least M units between or equal to every pair of quantiles.

Precision Measures

The precision measures of an estimate (e.g., standard error, confidence interval) and hypothesis tests are not releasable if the estimate itself is not releasable. In the case of the desire to have multiple precision measures released for a single estimate (e.g., the standard error, confidence interval, and C.V. of an estimate), more than one measure can be released if they are each calculated based on rounded estimates. Otherwise, one exact precision measure can be released but the others must be based on rounded estimates. In addition, confidence intervals must be centered around the rounded estimate, not the exact estimate, to prevent the use of the standard error to reveal the exact estimate. Hypothesis test results (e.g., chi-square or a model F-test) are releasable unrounded; any degrees of freedom that are reflective of the sample size must be rounded to the appropriate base.

Global F-tests and overall model statistics are not releasable if the entire model does not pass sensitivity testing.

Suppression

Suppression is not permitted in the RDCs / FRDC as a technique to deal with values / cells that do not meet vetting rules. Suppression is the process of removing additional (secondary) values from an output to mask the presence of an otherwise unreleasable value. For example, descriptive tables, with cells whose values which do not meet the rules, must be redesigned to meet the rules before they can be released. Suppression does not adequately protect against residual and attribute disclosure (discussed below) with descriptive outputs.

Residual Disclosure

Residual disclosure refers to the situation where an observation is potentially identifiable due to requesting two sets of similar outputs. These outputs could be in the same vetting request, or in separate vetting requests (for example, if a reviewer's comments require a table to be respecified). As

adjustments to the sample will likely occur over the course of the project, residual disclosure issues may prevent Statistics Canada from releasing the adjusted analytical output in later requests. Residual disclosure is most likely to occur when dealing with simple statistics over the course of the project, but it can occur with model output as well. When there is a residual issue in an analytical output being requested for vetting, the output must be modified so that no residual issue exists.

With respect to tabular output, residual issues can arise in several situations.

(A) Changes to Variable Coding / Changes to Filter or Stratification Variables

For models sample sizes and descriptive outputs (e.g., by employee status, business size), residual disclosure can occur from slight changes to the analytical approach (e.g., recoding variables, changing filters / stratification variables). Residual situations can also occur with the combination of several tables using the same variables in the same dataset but with different filters that makes it possible to identify a set of units or produces a “shadow table” that contains counts that are not above the threshold for release.

(B) Suppression of a Single Category that has Valid Counts but Does Not Meet the Threshold for Release

In some cases, data users may, for a particular analysis, desire to omit some valid cases from that analysis because the counts for a particular variable category are not releasable. For example, if using Census Subdivisions (CSD) in a descriptive table, small CSDs may be filtered out of the analysis for that particular descriptive table but may be used elsewhere, such as in a regression model. However, the valid CSDs included with the total sample size from the regression model can be used to recreate the missing CSD counts from the descriptive table. As suppression of unreleasable counts is not permitted and data users should consider using the same variable classifications throughout the analysis so that the analysis sample stays the same across all outputs and that all minimum counts are met.

(C) Exclusion of Cases and Missing Data

Missing values that are due to non-report (e.g., Don't Know, Not Answered) do not have to be checked for meeting the minimum count thresholds if they are removed from the data prior to running analyses on the data. If these counts are requested for vetting, they are treated as valid observations and all applicable vetting rules apply. Exceptions are the “Valid Skip” and “Not Applicable” responses – these can identify contributing units when used in combination with responses from other variables and must be checked regardless of if they are part of the vetting request or not.

Cases which are excluded from a particular analysis (e.g., through a filter variable) must still be vetted as if they are a dimension in the analysis to protect against residual disclosure. Generally, this is only an issue if the excluded population does not meet the threshold for release. For example: Vetting a table about the population with employment income will require also vetting the population without employment income in the same analysis sample if there is the possibility of revealing the total population in the sample.

(D) Previously Released Output

It is recommended to leave the vetting of frequencies and descriptives until the very end of the project to avoid residual disclosure risks. Any analytical output that is submitted for vetting is checked against all previous analytical output that has been released for the project. A request for output that creates a residual from any previously released output will result in that request not being released. It is

recommended that data users keep track of their own vetting requests so that residual confidentiality issues can be avoided, and also to avoid requesting the same output multiple times.

Longitudinal Analyses

While made explicit where relevant, in most cases, for analyses spanning multiple months or years, **the vetting rules must be applied to each period used in the analysis in order for the output of that analysis to be released as a whole**. For example, a regression model with “year of business opening” as the only independent variable and entered the model as a series of dummy variables would require the output for each year to pass all applicable vetting guidelines. If multiple years are pooled and analyzed as a single dataset, then the separate parts of time do not need to be tested separately (i.e., vetting is done on the full analytical sample). However, data users should be aware that number of unique units across the entire time span may not correspond to the model sample size due to business entities showing up in multiple records (see the section on [Minimum Counts / Sample Sizes](#) above).

Custom Aggregations of Variables

Any custom aggregation of variables that differ from standard group classifications (e.g., geographies) should be clearly indicated in both the output provided for vetting and in the supporting documentation provided to the Analyst. Custom combinations of categories may be problematic if unique combinations are created that lead to the identification of an unreleasable count or estimate through a residual process. For example, releasing output for both British Columbia combined with PEI and for British Columbia alone will also reveal the same output for PEI through a residual.

Large households (or large economic families) are easily identifiable. Therefore, in tabular output, variables related to household sizes should be capped at 5 people and the records should be collapsed for all households with 5 or more people into a single category (e.g., “5 or more people”).

Graphs, Figures, and Maps

Graphs can represent data in many ways – as individual data points, as a plot of means or model coefficients, or as predicted values. Graphs such as scatterplots and residual plots which show individual data points are not allowed for release. For graphs that involve geographical areas, area suppression is not allowed. Collapsing or other redesign techniques must be used instead of dropping problematic geographic areas. Any graph being considered for release should be discussed with the Analyst.

Sensitivity Testing

To minimize the risk of disclosure for business data, the notion of sensitivity testing of analytical outputs is central to the vetting approach. For economic data in the RDCs / FRDC sensitivity testing incorporates two concepts: minimum counts and dominance.

Minimum Counts

Minimum counts have been [discussed previously](#) in this document.

Dominance

A common problem with aggregates from economic surveys is that of dominance. Dominance occurs when one or a few businesses account for most or all of a cell total.

Sensitivity Rules

Several sensitivity rules are used to identify variables with dominance or influence that are a risk for confidentiality.

1. **Ambiguity rule, also called p-percent rule:** A cell is deemed sensitive (confidential) if the sum of all but the largest two respondents account for less than p-percent of the largest respondent's value. For example, with a p-percent rule of 30% where the largest respondent contributes a value of 100,000 the cell is deemed sensitive if the sum of all but the largest two values is less than 30% of 100,000 (i.e., smaller than 30,000).
2. **Dominance rule, also called (n,k) rule.** A cell is sensitive if the largest n contributors account for more than k% of the cell total value. For example, with a (6,80) rule a cell is sensitive if its largest 6 contributors account for more than 80% of the total for that cell.



In the RDCs/FRDCs a combination of the p-percent and dominance rules (known as the C2 rule) is used for economic data. The actual parameters for these rules are confidential and are not to be shared with data users.

Sensitivity Testing of Variables

Sensitivity testing is done on all continuous variables. Discrete variables that **are not specified as continuous** should also have sensitivity tests conducted on them. Categorical variables that act as “flag” variables or are dichotomous to indicate presence/absence of a characteristic are not subject to sensitivity testing. For descriptive statistics or saturated regression models, if a statistic is being released stratified by a categorical variable, test for sensitivity of the continuous variable at each level of the categorical variable.

Variables that are transformed (e.g., logarithmic) have sensitivity tests done on the original untransformed variables.

The Data Access and Dissemination (DAD) Division has created a confidential user-friendly SAS Sensitivity macro which is provided in the RDCs / FRDC for the purposes of identifying variables in the analytical output of a vetting request that are considered sensitive. This macro has options for reading data sets in SAS, SPSS, and STATA formats and extensive knowledge of SAS is not required to use the macro.

In the case of variables being found to be sensitive, data users are required to recode variables or restructure analyses so that the sensitivity issue is no longer present. **Suppression of a sensitive variable or of a table entry for a sensitive variable is not allowed (an exception for model outputs is described below).** The results of the sensitivity tests should be provided to the Analyst as supporting documentation and should be clearly labeled so that the Analyst can associate the sensitivity test results with the proper output. More information about conducting the sensitivity tests is given in the section on using the [Sensitivity Macro](#).

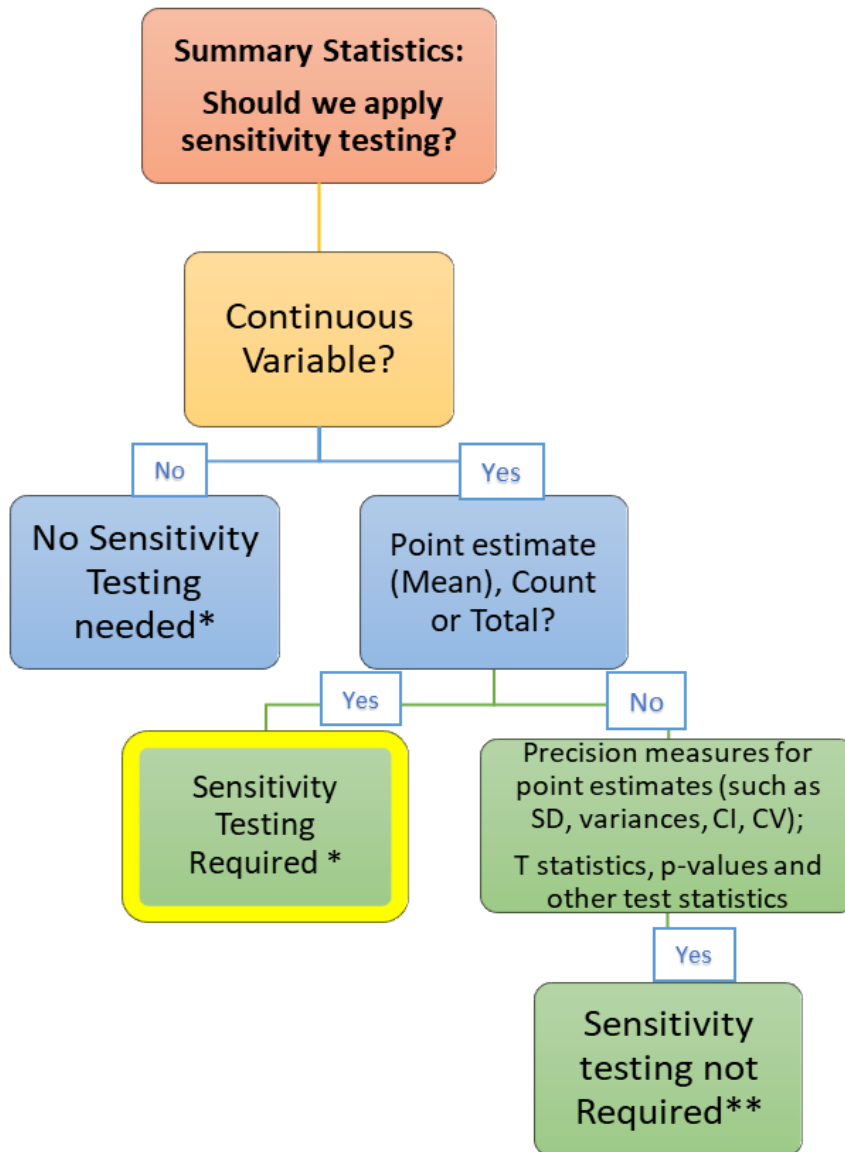
Precision measures such as a standard error or a standard deviation must also be checked for sensitivity prior to being released by assessing the sensitivity of the variable in the same manner as done with assessment of means/totals.

Vetting Considerations for Counts and Summary Statistics

Many of the points discussed previously in this document pertain to the vetting of counts and summary statistics, which will only be released in limited circumstances. The table below provides a summary of the issues that should be considered by data users for these types of outputs. All descriptive output requested for vetting must be shown to be (a) part of the approved proposal for access, (b) not disallowed by Subject Matter in the proposal evaluation, and/or (c) supporting model output. Any descriptive output that does not meet these criteria cannot be released until an amendment detailing the need for the additional descriptive outputs has been approved.

	Vetting Summary for Summary Statistics			
	Minimum Counts	Rounding	Sensitivity Testing	Notes
Frequency (count)	Unique contributing units	Yes	No	Must be high-level tabulations to be released
Proportion (of counts)	Unique contributing units Numerator, denominator, and their difference	Yes	No	Must be at a high-level to be released
Mean & Total	Unique contributing units	No	Yes	Must be at a high-level in order to be released
Quantile	Unique contributing units Below and at cutpoint Above and at cutpoint	No	No	Minimum counts of unique contributing units between each pair of requested quantiles Must be high-level descriptives in order to be released
Correlation of Continuous Variables	Unique contributing units Total counts	No	No	If the correlation involves a dichotomous variable then counts at each level of the dichotomous variable are required

Decision Tree for applying Sensitivity Tests to summary statistics with Economic Data



* Only continuous variables need to be tested for sensitivity, stratified by any categorical variables

** Precision measures, T statistics, p-values and other test statistics are releasable only if the point estimate is releasable so ensure the point estimate is sensitivity tested.

Vetting Considerations for Model-Based Outputs

All models must be vetted. There are situations where models are equivalent to simple counts or descriptive statistics (also known as “table equivalent models”); for example, a regression model such as

$$\text{Revenue} = (\text{constant}) + \text{beta} (\text{firm_size}) + \text{error}$$

produces output which is equivalent to having the mean of revenue for each level of firm size (or the counts / proportions of firm size in a logistic regression model). In the output below, firm size is coded as large=1 and small =0, and the sum of the constant and the unstandardized beta coefficient give the mean of the dependent variable for where firm_size=1 (large).

Descriptives			Model Output	
	Counts	Mean on DV	Constant	13.94
Large = 1	667	13.91	Beta for firm_size	-0.03
Small = 0	875	13.88		
			$13.945 + (-0.03) = 13.91$	$13.94 + (2 * -0.03) = 13.88$

Moreover, many programs provide model diagnostic information such as R^2 values, variance inflation factors (VIF) and residual tables or plots that are used to evaluate the adequacy of the model.

The following types of regression models or associated model output cannot be released:

- (1) Table-equivalent regression models: Examples of these models that have a single independent variable which is dichotomous, or a single independent variable represented by dummy categories, or has multiple independent variables but every main effect and interaction effect is specified.
- (2) Models that can be collapsed to a graph or summary statistics.
- (3) Model with near-perfect prediction (these models can be combined with public information that can reveal information about a particular business entity; these should be evaluated with the Analyst).
- (4) Model-predicted values and/or residuals at the unit level.
- (5) Variance-covariance matrices of model coefficients.
- (6) Model diagnostics cannot be released unless the entire model passes sensitivity tests.

It is strongly recommended that data users not ask for residual information to be vetted in numerical form; ideally a verbal description of the results from the diagnostics would be sufficient.



Data users should discuss any models that have a single independent variable or interaction effects with their Analyst early in the research process.

Sample Sizes from Models

Many software packages provide the unweighted sample size for models as part of the output. These model sample sizes must be vetted with care as the sample sizes from several models could be used to

generate a descriptive table. Further, models must satisfy the minimum count threshold for the number of unique analytical units, which may be different from the sample size of the model.

Sensitivity Testing with Regression Models

Every regression model must go through sensitivity testing for:

1. All continuous variables in the model (both dependent and independent),
2. If the model is a series of only dummy variables, test a continuous dependent variable stratified by each dummy variable.

If a researcher requests only test statistics and p-values associated with each of the coefficients for continuous variables, the sensitivity tests must still be conducted on those variables.

If any of these tests fail, then the model as a whole is not releasable. For multivariate models, a variable which does not pass the tests can be kept in the model as a control variable; however, any information about that variable (e.g., coefficient, hypothesis test results) and the model constant cannot be released.

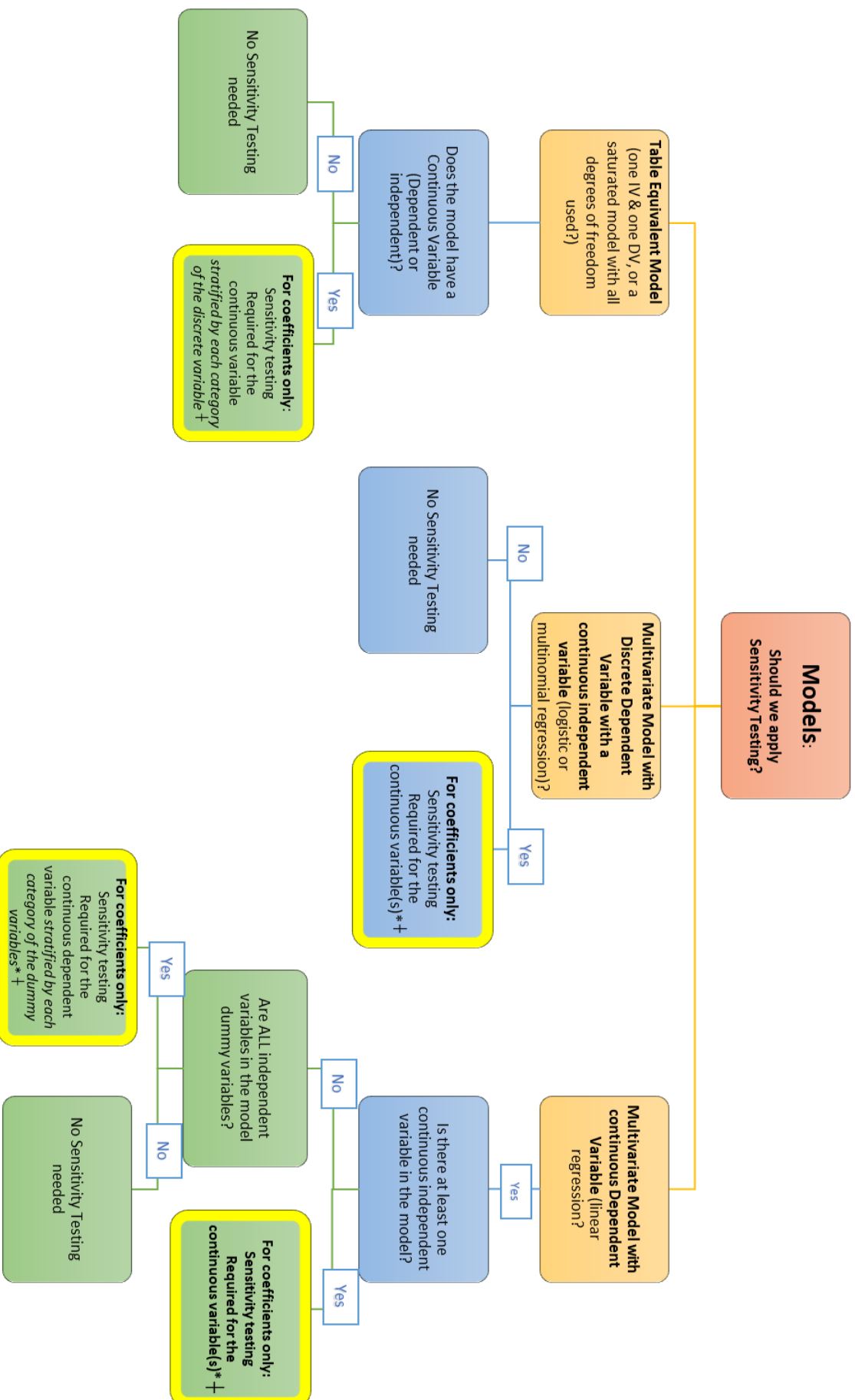
Since the sensitivity tests rely on an accurate sample size and the sample size can change from model to model depending on missing data or selection criteria, the specific analytic sample for each model must be saved and used for the corresponding sensitivity testing. For example, if the total sample size is 150 but a model with 3 independent variables only has an effective sample size of 100, then the sensitivity testing must be done on that 100-respondent sample size and not the total sample. Below is a table with some suggestions for saving an analysis-specific dataset in various software packages.

Checklist for Vetting Models

The following table outlines the vetting checks that need to be performed for various types of variable combinations. If the specific modeling approach being used is not listed here then please discuss with the Analyst.

Saving an analysis-specific dataset in STATA, SPSS, SAS, and R	
STATA	Use the “gen” and “e(sample)” statements after an analysis to create a flagging variable that can be used to identify which cases correspond to which analysis. reg y x1 gen flag_output1 = e(sample) * This creates a variable that is “1” if included in the analysis
SAS and SPSS	Create a flag variable to represent which cases in a dataset are involved in an analysis (e.g., model1=0 for not included, model1=1 for included) and save this dataset as a new dataset. In SPSS the DATA -> Select Cases option can be used to remove cases that do not meet a particular criteria, and then the resulting dataset can be saved.
R	The following code can be used to create a new dataset in R: Model1 <- “specific regression model command” Dataset[rownames(Model1\$model),]\$flag=1 If an R-package is being used in the analysis, data users should check if the package can allow for the model to be stored in order to save the analysis-specific dataset.

Decision Tree for applying Sensitivity Tests to Models with Economic Data



*** For multivariate non-table equivalent models:** Only the continuous variables need to be tested. They do not need to be stratified by the discrete variable categories, **unless** the model independent variables are all dummy variables. If all independent variables are dummy variables, the sensitivity testing must check the continuous dependent variable stratified by each dummy variable

+ Hypothesis test parameters are releasable if the coefficient is releasable

Sensitivity Macro for Economic Data

The Data Access and Dissemination (DAD) Division has developed an easy-to-use computer macro to aid data users in with checking some of the vetting rules for economic data. The “sensitivity macro for economic data” program (hereafter “sensitivity macro”) is a confidential SAS program designed to check for dominance, influential observations, and minimum count thresholds of each analytical output to prevent the disclosure of information about a business. The sensitivity macro uses a set of confidential rules developed at Statistics Canada. All output requested for vetting must have any dimension / filter / stratification variables, model independent variables, and all outcome variables checked for sensitivity when applicable.

Data Setup, Macro Preparation, and Output Interpretation

While the sensitivity macro is written in the SAS language, **it is not required to have previous knowledge of SAS programming in order to use the sensitivity macro**. The sensitivity macro can accept data sets from SPSS and STATA, and much of the code has been pre-written; data users will have to know the **names of the variables** involved in the analysis, the **folder paths to the data**, and the **folder path for the results** of the sensitivity macro.

The sensitivity macro requires 6 inputs: (1) the microdata file, (2) the variable that identifies the unique data units/businesses, (3) the outcome variables in the microdata to be checked, (4) the dimension specifications, (5), the minimum count threshold for the dataset being analyzed, and (6) the destination for the macro output files which will contain the results of the sensitivity tests. Prior to running the macro the variables should be checked at each dimension level to ensure that the minimum counts are met. STATA users should ensure that string variables have been converted to numeric format, and should use the “saveold” command to save the dataset going into the macro.

The output table of the macro will report the cells of totals or aggregated counts for the outcome variable over the levels of the dimension variables specified in the macro. Since the macro counts the number of unique data units/businesses, the sample size reported by the macro may be different from what is seen in the analytical outputs. If the number of units reported by the sensitivity macro is very low (and too low to be the number of unique respondents or businesses), data users should check that the respondent identification variable is specified correctly in the macro.

Examples of how to use the sensitivity macro syntax and associated outputs are provided in [Appendix A](#).

Due to the requirement for multiple data sets to be created for the purpose of sensitivity testing, server space limitations will prevent data users from saving multiple versions of the data for each piece of analytical output. It is strongly recommended that flag or filter variables be created in the data to select the appropriate subsample to be analyzed for each piece of the analytical output and that these flags be used in the macro syntax.

Vetting Requests

What to Provide to the Analyst for Vetting

The following should be provided to the Analyst with each analytical output to be vetted:

- The syntax that was used to create the analysis data, the syntax for each analytical output being vetted, and the syntax that was used for the sensitivity macro (data users should make it clear which syntax applies to which output)
- All supporting documentation (output from the sensitivity macro, crosstabulations or counts for models) as required
- If the output is in response to a journal review, then the review should be provided along with an indication of which of the reviewer points are being addressed in the output

Vetting Output from Extensions / Revisions of an Ongoing Contract

Extensions are a continuation of a project to a new end date, and revision contracts are new contracts (opened for the purpose of addressing reviewer comments). In both cases, for the purposes of vetting, extension and revision contracts are treated as a continuation of the original project. As a result, the vetting of output is checked against the original project and previous vetting to ensure against residual disclosure issues.

Warning Signs to Be Aware Of

Linear regression – ratio of sample size to number of estimated parameters

If a linear regression model is estimated on a small number of units, then releasing the estimated parameters is more likely to lead to a disclosure. The sample size (n) should be much higher than the number of estimated parameters (p) – a sample size of $10 \times p$ should be sufficient if all applicable vetting rules are met.

Table-equivalent models

If the regression model is fully saturated, the estimated coefficients will reflect a table of means. These models are not releasable as they are equivalent to summary statistics. If a binomial or multinomial regression consists solely of categorical independent variables that are dummies, then the model is equivalent to a crosstabulation between the dependent variable and the single categorical variable comprising the dummies.

Nested models

If separate regressions are performed but with the only difference being the change of a few units into the analysis sample or an independent variable, then significant changes in the estimated model parameters could lead to a residual disclosure. For example, if adding one observation in the model produces large changes in the estimated coefficients, these changes can be attributed to this new observation.

Disclosure by repeated differencing / Nested Models

If a data user estimates the same model twice, with additional observations to show the effect of problematic observations, this could lead to several scenarios that could be problematic for confidentiality.

Scenario 1: If the data user knows the number of observations from the original model, it is possible to solve the model and disclose the information of all the observations in the model.

Scenario 2: If the data user does not know the original number of observations for the model, the estimated variance-covariance matrix can be used to reproduce the cross-product matrix. Then, if the new observations are known (e.g., from public sources) there is now sufficient information to reproduce the data for these added observations.

Scenario 3: As the regression line goes through the mean of the variables, the mean value of the dependent variable from the new observations can be identified if the means of the predictors from both models are available. If there is only one additional observation, this discloses the value of the additional dependent variable.

It should be noted that disclosure of the mean value of the dependent variable based on the new observations does not require disclosure of the mean value of the dependent variable from the original set of observations.

Scenarios 1 and 2 are feasible in the secure access points (SAPs). Running the same regression on a subset of the data is an accepted way of analysing the impact of sampling. Both scenarios require substantial additional knowledge, and all other observations or the variance-covariance matrix (which is unsafe and so not automatically released in controlled environments, and unlikely to be published) must be known.

Scenario 3 is the most realistic situation, where the means of the two variable sets used in each model are known; this is possible as means are often published in journal articles. This enables the means of the additional observations to be identified and, as a result, means are an “unsafe” statistic since disclosure is non-negligible when combined with the regression. In summary, the likelihood of disclosure depends upon some very specific scenarios and, usually, some stringent information requirements to be known. Scenarios 2 and 3 also require repeated estimation under controlled circumstances. These conditions are not often fulfilled by genuine research activity, but data users should keep careful track of the models and sample sizes for each vetting request.

Frequently Asked Questions

1. “What do I do if a count/model does not meet the minimum sample size requirements” (or “I need this low cell count because YYY is my variable of interest”)

For counts that do not meet the minimum requirements, that count, and any associated descriptive / model output cannot be released (e.g., means, single-independent variable models). For tabulations that have cells that do not meet the minimum requirements, categories need to be collapsed together. For graphs (e.g., histograms, survival curves), binning may be required to allow the graph to be released.

Missing values (e.g., Don’t Know, Refusal) are generally not affected by the minimum sample size requirements as long as they are not requested for in a vetting request. Exceptions are the “Valid Skip” and “Not Applicable” responses – these can identify contributing units when used in combination with responses from other variables and must be checked regardless of if they are part of the vetting request or not.

It is recommended that data users consult with the Vetting Analyst and, if possible, the Vetting Analyst will make suggestions to remedy the situation. However, the Vetting Analyst must respect the vetting rules and sometimes the output cannot be released.

2. “I need all of this output so that my advisor/project leader can decide what they want to publish”

Large vetting requests or multiple requests that are the result of exploring the data and producing a lot of intermediate results for vetting must be avoided. If data users are not entirely sure of what output is needed it is advisable to add team members to the project so that everyone can see the output to determine what needs to be vetted. Another option is to have trends, or a summary note vetted (e.g., which variables are significant, if beta coefficients are positive or negative, which counts are low/medium/high) instead of numerical output.

3. I have an unexpected deadline, what can I do to help hurry the output release?

It is recommended that data users meet with the Analyst and ask for only the absolute necessary output to be released. For example, instead of asking to release a complete set of outputs, data users can bring

in a set of blank tables that would be used to meet the deadline and ask to release those tables only (as a supporting document the raw output that produced the values for those tables would have to be provided). Data users can return later to ask for more output to be vetted.

4. How important is the release of the full set of model coefficients?

It is clear from the documentation presented above that the full set of estimated coefficients is necessary for vetting; therefore, any linear model which includes incidental (implicitly included but unpublished) parameters poses no disclosure risk. Any linear regression model where parameters contribute to the line of best fit but are not explicitly generated poses no disclosure risk under the scenarios described above. This usefully includes several classes of models, such as multilevel models.

5. Is statistical quality associated with risk?

The statistical quality of the data or output should not be confused with disclosure. Data problems leading to a highly skewed distribution can lead to more risk of approximate disclosure, but generally low quality data and poor models reduce disclosure risk.

6. What are the risks of releasing additional information?

These include the release of residuals, the variance-covariance matrix, minimal and maximal values, and quantiles. Residuals may be much more disclosive than regression coefficients. This is not a problem of regression coefficient risk, but tabulation risk. Residuals are microdata, albeit modelled, and so depictions of them are treated as “unsafe” statistics just as graphs, tabulations or distributions of source microdata would be.

Although formally an estimated value, the variance-covariance matrix is an “unsafe” statistic, as it can be used to derive the cross-product matrix on the reasonable assumption that the estimated error variance is published. It is “unsafe” because interactions with dummy variables can generate group means. However, the decision to publish regression coefficients does not require the publication of the variance-covariance matrix.

Conclusion

Given the range of techniques used in analysis, it is impossible for this document to be comprehensive and cover every possible type of output. Data users that are utilizing an analysis technique that is not discussed in this Handbook should meet with their Analyst to review and assess the technique and to determine the appropriate supporting output required for vetting. Any questions can be directed to the Vetting Analyst.

References

Tambay, J.L. (2009). Confidentiality of Tabular Data for Businesses. Internal document.

Tambay, J.L. (2009). Rules for Analytical Outputs involving Business data at RDCs. Internal document.

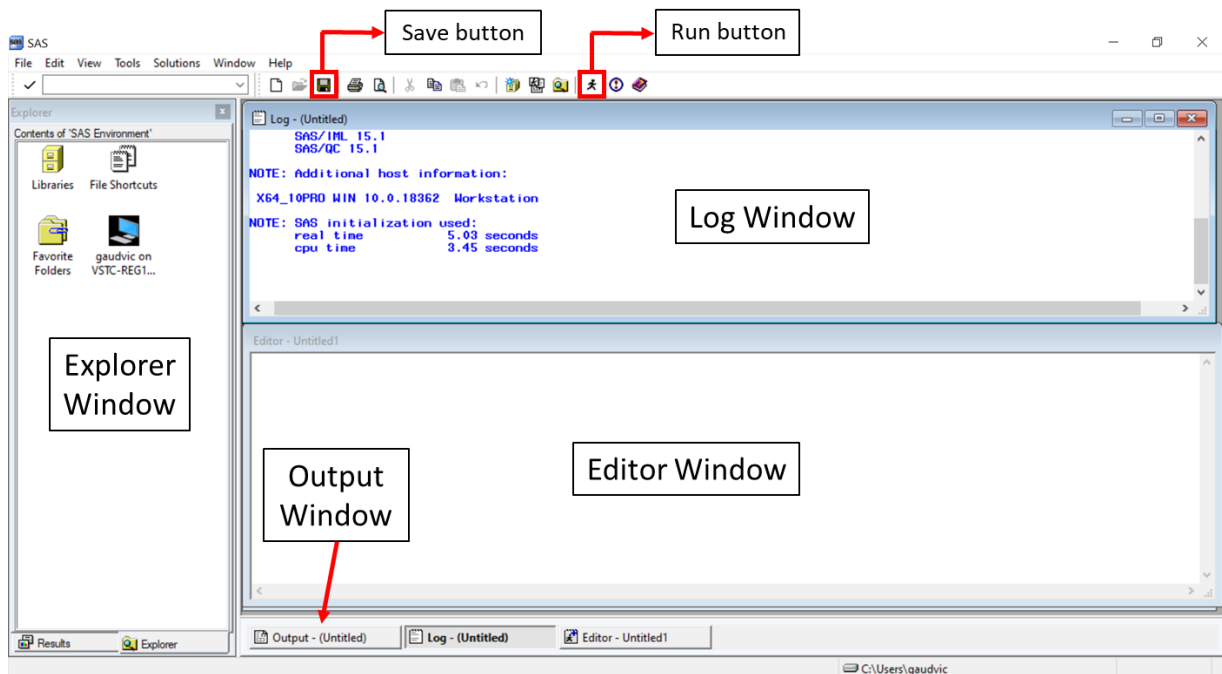
CDER (2017). Confidentiality Handbook for CDER. Internal document.

Appendix A – General Description of the Sensitivity Macro

The sensitivity macro program is designed to check for both sensitivity and dominance in table cells or a combination of cells of each analytical output to prevent the disclosure of information about a business. The sensitivity macro program uses a set of confidential rules developed at Statistics Canada which are not accessible to data users or SAP staff.

The SAS Environment

The SAS program provides users with a text-based interface, as seen in the image below.



The Sensitivity macro will be loaded into the Editor Window. The file location of the macro can be provided to data users by the RDC/FRDC Analyst.

Data Setup and Macro Preparation

The program requires the user to know 6 things:

- (1) The location on the server for the Sensitivity macro (provided by the RDC Analyst)
- (2) The data file used for the analysis output that is being checked by the sensitivity macro
- (3) The continuous variables in the analysis output to be checked for sensitivity
- (4) The dimension/-stratification variables used in the analysis
- (5) The minimum counts required
- (6) The destination folder for the sensitivity macro output files that will be provided as supporting documentation to the vetting request.

The microdata submitted to the sensitivity macro must have every variable of interest to be checked. The dimension variable(s) help create the hierarchy of aggregation for the variable in the microdata that the program analyzes. Prior to running the macro, the variables should be checked at each dimension level to ensure that the minimum cell counts are met.

Sensitivity Macro Syntax

The Sensitivity macro options are: **DATASET**, **ID** *variable*; **VARNAME**s *variable(s)*; **BYVAR**s *variable(s)*; **MINCOUNT** and **OUTFILE**.

Option	Content of the Option
DATASET=	Specify the name of the input microdata file that was used to produce the analytical table.
ID=	Specify the name of the ID variable in the microdata which is the unique respondent id
VARNAMEs=	Specify the name of the variable(s) in the microdata to be checked for sensitivity.
BYVARs=	Specify the name(s) of the variable(s) that represent the dimensions/stratifications.
MINCOUNT=	Specify the minimum count required
OUTFILE=	Specify the location of the output SAS data that will contain the results of the sensitivity tests

While the ID and BYVARs can be numeric or character variables, the VARNAMEs should be numeric. The BYVARs option is optional and can be left blank, but it is required when checking categorical variables for minimum counts or when used as stratification variables. In the absence of a variable in the BYVARs statement any variables will be treated as a continuous variable by the sensitivity macro.

The output of the sensitivity macro produces tables of the variable names that were provided in “VARNAMEs=” option. The tables will have columns with “fail_c2”, “fail_count” and the “fail_overall”. The “fail_c2” column is for checking the variable against the confidentiality parameters for sensitivity. The “fail_count” column is for checking the count of the unique number of analytic units against the minimum count threshold. The “fail_overall” will show a pass (the value of 0) if both the “fail_c2” and “fail_count” parameters pass, and will show a fail (a value of 1) if either the “fail_c2” or “fail_count” parameter fails. When the test is used to check for minimum count, the “fail_c2” column is not required; hence why the “fail_overall” column equals 1 (a fail) if *either* of the other two test columns fail. The required column to check is only the “fail_count” column, and any entries with a “1” indicate that the output must be modified prior to release.

Calling the Data in to the SAS Program

The following two SAS statements can be used to bring data in to SAS. The first uses a libname statement (this assigns a label to the folder that contains the data, and it can be used to easily refer to that location without typing out the path every time), and the second imports the data and saves it to the location specified in the libname statement:

```
libname input "(put in path to where the sensitivity data will be saved here)";  
proc import datafile="(put in path to data file here)" out= (label name).(dataset name)  
The libname "input" does not need to be explicitly used – data users can change this.
```

An example to read in the raw data from a T1FF business file and save it would be

```
libname tax1994 "C:\project1234\data";  
proc import datafile = "C:\project1234\original_data\T1FF_1994_business.sas;  
run;
```

If the data is in STATA format, all that changes is the proc import line:

```
Proc import datafile = "C:\project1234\original_data_\T1FF_1994_business.dta" dbms=dta replace;
```

If the data is in SPSS format, a similar change is made to the proc import line:

```
Proc import datafile = "C:\project1234\original_data_\T1FF_1994_business.sav" dbms=spss replace;
```

If data users are using a software package that can export the data directly to SAS format then these steps are not needed.

Calling the Sensitivity Macro

The following SAS code is used to call in the Sensitivity macro.

```
libname c2call "(path to where the macro is stored on the RDC/FRDC server)";  
options mstored sasmstore=c2call;  
libname out "(path to where the results from the sensitivity tests will be stored)";  
%c2rule (dataset=input.Data1, id=entid, varnames=revenue, byvars=years, mincount=M, outfile=out.);
```

The libnames for c2call and out can be changed by the data user. The Sensitivity macro will create output files in the location specified by the "out" libname which are given the filename of the variable in the "varnames" option of the %c2rule statement.

Examples of how to use the sensitivity macro for specific types of analyses, and the associated macro output, are given in the [Examples section](#).

Interpreting the Sensitivity Macro Output

The output table will report the sample counts for the continuous variable of interest over the categories given by the dimension/stratification variables entered in the byvars option of the macro. A "0" is given if an variable passes the sensitivity check, and a "1" if it does not pass. An example of what that looks like is given in the image below:

	years	revenue_N	fail_c2	fail_count	fail_overall
1	2000	20	0	0	0
2	2001	10	0	0	0
3	2002	10	0	0	0

The "fail_c2" column is for checking the varname variable (here, revenue), against the confidentiality parameters for sensitivity. The "fail_count" column is for checking the counts of the unique number of analytic units by each year (as specified in the byvars option) against the minimum count threshold. For continuous variables the "fail_overall" will

show a pass if both the “fail_c2” and “fail_count” parameters pass, and a fail if either of them fail. For categorical variables, where only the minimum count is being checked, the “fail_c2” column is not required and only the “fail_count” column is checked. **Any entries with a “1” indicate that the output must be modified and re-tested prior to being vetted for release.**

Supporting Documentation

Also, to be provided as part of supporting file is the data used for each analysis (or the syntax to create that data) In addition, the total observation counts of the variable(s) must be the same as the observation counts in the output tables requested to be released. If there are dimensions, a tabulation of the dimension variable(s) must be provided to the analyst and the sample sizes must align with the samples in the analytical outputs (if multiple dimensions are used then the appropriate crosstabulation table is required as supporting documentation). This check is necessary because if the IDs in the data are not unique, the sample sizes produced in the sensitivity macro tables will not be the same as those in researcher analytical output files that are to be released. As a result, a manual check of the sample sizes is required.

Examples of the Sensitivity Macro

To use the compiled macro in any other SAS session, data users will indicate the location of the compiled macro with the SAMSTORE option.

```
LIBNAME c2call '\\f4cder02\C2 test'; /*location of the compiled macro; adjusted by the
researcher*/
OPTIONS MSTORED SASMSTORE= c2call; /*options must have the libname above*/
```

Once the options are specified, the sensitivity macro can be run with the appropriate variables and minimum count threshold. An output location can be specified to save the output tables at that location using the “OUTFILE=” option.

```
libname output "\\f4cder01\VETTING\2015_XX_Project\Vetting\Sensitivity_Results";
/*Sensitivity Testing results saved at this location */
```

Example 1: An average of a continuous variable, “revenue” with a minimum cell count threshold of 5 units

Syntax:

```
%c2rule (dataset=input.Data, Id=entid, varnames =revenue, byvars=, mincount=5,
outfile=output.);
```

Output 1:

```
%c2rule (dataset=input.Data, Id=entid, varnames =revenue, byvars=, mincount=5,
outfile=output.);
```

	revenue_N	fail_c2	fail_count	fail_overall
1	39	0	0	0

The column “revenue_N” shows the unique number of analysis units (here, business enterprises) that are contributing to that mean. The “0”s in the other three columns show that the variable, “revenue”, passes all of the confidentiality guidelines. This output could be released.

Example 2: Averages of revenue by province

Syntax:

```
%c2rule (dataset=input.Data, Id=entid, varnames =revenue, byvars= province, mincount=5,
outfile=output.);
```

Output:

	Province	revenue_N	fail_c2	fail_count	fail_overall
1	Alberta	4	0	1	1
2	British_Columbia	4	1	1	1
3	Manitoba	4	0	1	1
4	New_Brunswick	4	0	1	1
5	Newfoundland_and_Labrador	4	0	1	1
6	Nova_Scotia	4	0	1	1
7	Ontario	4	0	1	1
8	Prince_Edward_Island	4	0	1	1
9	Quebec	4	0	1	1
10	Saskatchewan	4	0	1	1

In this example, for the variable “revenue”, all of the provinces except British Columbia have economic entities (here, enterprises) that are passing the sensitivity rules. However, all of the provinces show that they do not meet the minimum count threshold for unique entities, so this output would not be released.

It should be noted that the output tables produced by the macro will have the names provided in the “varnames” option so producing other tables with same variable names and same location will overwrite previously produced tables at that location.

Appendix B: DAD Confidentiality Vetting Request Form for Business Data

Research Data Centre Economic Data Confidentiality Vetting Request Form

Name:	Email:	Date:
Username:	Contract #:	
Project Title:		

Please check your output against the vetting guidelines; consult your RDC Analyst if you are unsure.

Summary statistics must be kept to a minimum and must support the analytical output.

Note: For students and research assistants, please have your supervisors and/or research team review the output before it is requested for release.

The completed Request Form is stored as part of the request record.

SECTION A. Checklist	Yes / No / NA
1) Is the requested output consistent with the approved proposal for this project?	Choose an item.
2) Have you subsetting or selected only a certain set of respondents from the data for all or part of the analysis? (e.g., males 50 years of age and older). If yes: a) Describe <u>each</u> of the different samples, sub-samples, or inclusions/exclusions used to produce your output in Section C . b) For each exclusion, specify whether it is for the entire project or specific to a sub-analysis in Section C .	Choose an item. Choose an item. Choose an item.
3) Have you checked the vetting rules to determine if there are geographical, institutional, business, and / or population requirements for your output?	Choose an item.
4) Variable construction a) For derived statistics and linear combinations of variables, provide supporting documentation showing each component used in the variable construction is releasable (i.e., minimum count and sensitivity). b) Non-standard aggregations by NAICS code, geography and firm employment size have been discussed with your analyst.	Choose an item. Choose an item.
5) Does this request include descriptive statistics? If yes: a) Do they support the model output? b) Clearly label output (tables have a title and every variable and category labelled). c) Ensure minimum counts and sensitivity are met as per the rules for the data. d) Provide correct supporting documentation according to the vetting rules (e.g., counts are unrounded and rounded).	Choose an item. Choose an item. Choose an item. Choose an item.

6) Does this request include model output or graphs that are equivalent to a descriptive statistic? (e.g., a model with a single independent variable, a model with all possible interactions, histograms). If yes: Please revise your model. Model output that is equivalent to a descriptive statistic cannot be released.	Choose an item.
7) Does this request include a correlation matrix? If yes: a) Provide the contributing unique units and sensitivity test for continuous variables. b) Provide the cross-tabulation of contributing units for dichotomous variables. c) For dichotomous variables correlated with a continuous variable, provide the contributing units and test for the continuous variable's sensitivity at each category.	Choose an item. Choose an item. Choose an item. Choose an item.
8) Rounding of output • Provide both the unrounded and rounded versions of the output. • Describe the approach to rounding and rounding base. • Ensure that any forced rounding to zero is clearly shown.	Choose an item. Choose an item. Choose an item. Choose an item.
9) Testing for minimum counts and sensitivity a) Provide a codebook in SECTION D . b) Check that the output (i.e., descriptive statistics, precision measures, and variables used in a model) meet the requirements for the minimum number of unique data units. c) Check all continuous variables and precision measures for sensitivity according to the vetting rules. d) Save documents related to sensitivity tests (e.g., SAS program, log files, output folders, etc.) in the supporting folder. e) Save and clearly label datasets used to create each output in the supporting folder.	Choose an item. Choose an item. Choose an item. Choose an item. Choose an item.
10) Is the requested output your <u>final</u> output? a) If no – future vetting release requests under this contract may be restricted due to residual disclosure. You are strongly encouraged to consult with your Analyst. Economic projects are expected to have limited intermediary output. Due to residual disclosure , the output requested has to be as close to final as possible. Multiple iterations of the output will not be released , barring an academic/journal review has been requested for the project. Consult with your Analyst for strategies to release intermediary output.	Choose an item.

SECTION B. Residual disclosure risk: Comparison with other output for this project	Yes/No
1) Have you any subsetted variables, where one or more variables is a subset of another? (e.g., had depression in the past 5 years and had depression in the previous year; had a chronic condition and had a respiratory chronic condition)	Choose an item.
2) Has a version of this output, in part or in whole, been previously released? If no, skip to Section C . If yes, compared to other outputs for this project have you: a. Changed the sub-sample or population of interest? b. Dropped individual cases or outliers?	Choose an item. Choose an item. Choose an item.

c. Imputed the missing values?	Choose an item.
d. Recoded or modified any variables even slightly?	Choose an item.
If the answer is YES to any of the questions above, discuss with your Analyst and provide the following supporting documentation:	
a. Explanation of changes.	Choose an item.
b. Description of how the output relates to other output for this project.	Choose an item.
c. Date(s) of the previous vetting requests related to this request.	Choose an item.
d. Residual tables as supporting files. Please refer to the Vetting Orientation.	Choose an item.
e. Provide both sets of syntax and highlight or indicate the changes.	Choose an item.

SECTION C. Output Requested for Release

Clearly label output and delete any output or values you do not want, or need released at this time.

File Name (list each sheet for spreadsheet files)	Source dataset(s) and cycle(s)/time period	Weight variable used (indicate if scaled or normalized)	Method used from list below	Sample Description (e.g., "employed females, aged 21-45, in Ontario")	Level of geography of requested output
			Choose an item.		Choose an item.
			Choose an item.		Choose an item.
			Choose an item.		Choose an item.
			Choose an item.		Choose an item.
...			Choose an item.		Choose an item.

Types of Methods

1. Descriptive (e.g., frequencies, cross-tabular analysis, means, distributions, ANOVA)
2. Scaling (e.g., factor analysis)
3. Graphs (e.g., histograms) – please remember to include supporting tabulations
4. Multivariate regression analysis (e.g., OLS, logistic, probit, tobit, poisson)
5. Complex modeling (e.g., structural equation modeling, hierarchical linear modeling, growth analysis, survival analysis, event history analysis, simultaneous-equations models, fixed effects models, random effects models)
6. Other – please describe (e.g., notes and syntax files)

SECTION D. Supporting files

These files are to support the vetting request and **will not be released**.

Please name your support files to allow easy pairing of the corresponding output file.

Place these files in your Supporting Documents folder

- 2) Syntax (or log) files for analysis
- 3) Supporting frequencies (e.g., unrounded output)
- 4) Supporting documentation for derived or recoded variables (i.e., syntax, codebook, description)
- 5) Supporting documentation for sensitivity tests
- 6) If applicable, supporting documentation for residual disclosure risk.

File name	Notes
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Place these files in your Supporting Documents folder

- 2) Syntax (or log) files for analysis
- 3) Supporting frequencies (e.g., unrounded output)
- 4) Supporting documentation for derived or recoded variables (i.e., syntax, codebook, description)
- 5) Supporting documentation for sensitivity tests
- 6) If applicable, supporting documentation for residual disclosure risk.

List of variables

Please list all variables used to generate the above outputs or paste a file path to your codebook here:

Please identify the weights used in row one.

	Variable Name	Associated Programs and formula	Type of variable	Derived
1			Choose an item.	Choose an item.
2			Choose an item.	Choose an item.
3			Choose an item.	Choose an item.
4			Choose an item.	Choose an item.
...			Choose an item.	Choose an item.

SECTION E. Additional comments which may be helpful to the analyst: